spark

```
Activities  Terminal                                   Mar 30 15:46
                                  student@student: ~/output
student@student:~$ echo $SHELL
/bin/bash
student@student:~$ nano ~/.bashrc
student@student:~$
student@student:~$ source ~/.bashrc
student@student:~$ echo $PATH
/home/student/DSBDAL/spark-3.5.1-bin-hadoop3/bin:/home/student/.local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/snap/bin
student@student:~$ spark-shell
26/03/30 15:20:28 WARN Utils: Your hostname, student resolves to a loopback address: 127.0.1.1; using 10.11.5.183 instead (on interface enp3s0f1)
26/03/30 15:20:28 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/03/30 15:20:31 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Spark context Web UI available at http://10.11.5.183:4040
Spark context available as 'sc' (master = local[*], app id = local-1774864232212).
Spark session available as 'spark'.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.5.1
      /_/

Using Scala version 2.12.18 (OpenJDK 64-Bit Server VM, Java 21.0.10)
Type in expressions to have them evaluated.
Type :help for more information.

scala> 26/03/30 15:20:44 WARN GarbageCollectionMetrics: To enable non-built-in garbage collector(s) List(G1 Concurrent GC), users should configure it(them) to spark.eventLog.gcMetrics.youngGenerationGarba
geCollectors or spark.eventLog.gcMetrics.oldGenerationGarbageCollectors
val inputfile = sc.textFile(&quot;/home/student/DSBDAL/input.txt&quot;)
<console>:1: error: ')' expected but ';' found.
       val inputfile = sc.textFile(&quot;/home/student/DSBDAL/input.txt&quot;)
                                                                          ^
<console>:1: error: ';' expected but '.' found.
       val inputfile = sc.textFile(&quot;/home/student/DSBDAL/input.txt&quot;)
                                                                          ^
<console>:1: error: illegal start of definition
       val inputfile = sc.textFile(&quot;/home/student/DSBDAL/input.txt&quot;)
                                                                          ^

scala> val inputfile = sc.textFile("/home/student/DSBDAL/input.txt")
inputfile: org.apache.spark.rdd.RDD[String] = /home/student/DSBDAL/input.txt MapPartitionsRDD[1] at textFile at <console>:23

scala> val counts = inputfile.flatMap(line =&gt; line.split(&quot; &quot;)).map(word =&gt; (word,
<console>:1: error: ')' expected but ';' found.
       val counts = inputfile.flatMap(line =&gt; line.split(&quot; &quot;)).map(word =&gt; (word,
                                                                  ^
<console>:1: error: ')' expected but ';' found.
       val counts = inputfile.flatMap(line =&gt; line.split(&quot; &quot;)).map(word =&gt; (word,
                                                                  ^
<console>:1: error: illegal start of definition
       val counts = inputfile.flatMap(line =&gt; line.split(&quot; &quot;)).map(word =&gt; (word,
                                                                  ^

scala> val counts=inputfile.flatMap(line=>line.split(" ")).map(word=>(word,1)).reduceByKey(_+_);
counts: org.apache.spark.rdd.RDD[(String, Int)] = ShuffledRDD[4] at reduceByKey at <console>:23
```

```
Activities  Terminal                                   Mar 31 12:07
                                  student@student: ~
student@student: ~                                student@student: ~/TCC48Ass
 GNU nano 6.2                                  /home/student/.bashrc
*)
    ;;
esac

# enable color support of ls and also add handy aliases
if [ -x /usr/bin/dircolors ]; then
    test -r ~/.dircolors && eval "$(dircolors -b ~/.dircolors)" || eval "$(dircolors -b)"
    alias ls='ls --color=auto'
    #alias dir='dir --color=auto'
    #alias vdir='vdir --color=auto'

    alias grep='grep --color=auto'
    alias fgrep='fgrep --color=auto'
    alias egrep='egrep --color=auto'
fi

# colored GCC warnings and errors
#export GCC_COLORS='error=01;31:warning=01;35:note=01;36:caret=01;32:locus=01:quote=01'

# some more ls aliases
alias ll='ls -alF'
alias la='ls -A'
alias l='ls -CF'

# Add an "alert" alias for long running commands.  Use like so:
#   sleep 10; alert
alias alert='notify-send --urgency=low -i "$([ $? = 0 ] && echo terminal || echo error)" "$(history|tail -n1|sed -e '\''s/^\s*[0-9]\+\s*//;s/[;&|]\s*alert$//'\'')"'

# Alias definitions.
# You may want to put all your additions into a separate file like
# ~/.bash_aliases, instead of adding them here directly.
# See /usr/share/doc/bash-doc/examples in the bash-doc package.

if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi
export PATH="/home/student/TCC52/spark-3.5.1-bin-hadoop3/bin:$PATH"
                                          [ Wrote 118 lines ]
^G Help    ^O Write Out  ^W Where Is   ^K Cut     ^T Execute   ^C Location  M-U Undo    M-A Set Mark  M-] To Bracket  M-Q Previous  ^B Back      ^← Prev Word
^X Exit    ^R Read File  ^\ Replace    ^U Paste   ^J Justify   ^/ Go To Line M-E Redo    M-6 Copy      ^Q Where Was    M-W Next      ^F Forward   ^→ Next Word
```

```
Activities   Terminal                                   Mar 31 12:10
                                                        student@student: ~
student@student: ~                                      student@student: ~/TCC48Ass

student@student:~$ spark-shell
bash: /home/student/DSBDAL/spark-3.5.1-bin-hadoop3/bin/spark-shell: No such file or directory
student@student:~$ cd
student@student:~$ ls
 1.java                    assignment3.java   Downloads           glfw-3.3.8                  lp11.java      'practical 8.ipynb'   TCC48Ass          Untitled1.ipynb   Untitled9.ipynb
 2.java                    assignment3.txt    DSDBlab1.ipynb      glfw-3.3.8.zip              LRU.class      pract.py              Templates         Untitled2.ipynb   Untitled.ipynb
 ari.java                  BCB53.ipynb        eclipse-workspace   iris.csv                    LRU.java       Public                uber.csv          Untitled3.ipynb   Videos
 ArithmeticOperations.c    BE03.py            emails.csv          libArithmeticOperations.so  Music          pythonlib             Untitled10.ipynb  Untitled4.ipynb
 ArithmeticOperations.class Boston.ipynb      fcfs.java           libAritmeticOperations.so   output         sales_data_sample.csv Untitled11.ipynb  Untitled5.ipynb
 ArithmeticOperations.h    DB.ipynb           FIFO.class          libgraph-1.0.2              Pictures       snap                  Untitled12.ipynb  Untitled6.ipynb
 ArithmeticOperations.java Desktop            fifo.java           libgraph-1.0.2.tar.gz       pract7.class   Social_Network_Ads.csv Untitled13.ipynb Untitled7.ipynb
 ArithOps.java             Documents          FIFO.java           loan_data.csv               pract7.java    student_data.csv      Untitled14.ipynb  Untitled8.ipynb
student@student:~$ nano ~/.bashrc
student@student:~$ source ~/.bashrc
student@student:~$ spark-shell
26/03/31 12:10:40 WARN Utils: Your hostname, student resolves to a loopback address: 127.0.1.1; using 10.11.5.202 instead (on interface enp3s0f1)
26/03/31 12:10:40 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/03/31 12:10:43 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/03/31 12:10:43 WARN Utils: Service 'SparkUI' could not bind on port 4040. Attempting port 4041.
Spark context Web UI available at http://10.11.5.202:4041
Spark context available as 'sc' (master = local[*], app id = local-1774939243575).
Spark session available as 'spark'.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.5.1
      /_/

Using Scala version 2.12.18 (OpenJDK 64-Bit Server VM, Java 21.0.10)
Type in expressions to have them evaluated.
Type :help for more information.

scala>
```

```
scala> val inputfile = sc.textFile("/home/student/tcc52/input.txt")
inputfile: org.apache.spark.rdd.RDD[String] = /home/student/tcc52/inputtcc52 .txt MapPartitionsRDD[9] at textFile at <console>:23

scala> val counts=inputfile.flatMap(line=>line.split(" ")).map(word=>(word,1)).reduceByKey(_+_);
counts: org.apache.spark.rdd.RDD[(String, Int)] = ShuffledRDD[12] at reduceByKey at <console>:23

scala> counts.toDebugString
res1: String =
(2) ShuffledRDD[12] at reduceByKey at <console>:23 []
 +-(2) MapPartitionsRDD[11] at map at <console>:23 []
    |  MapPartitionsRDD[10] at flatMap at <console>:23 []
    |  /home/student/TCC48/tcc48input.txt MapPartitionsRDD[9] at textFile at <console>:23 []
    |  /home/student/TCC48/tcc48input.txt HadoopRDD[8] at textFile at <console>:23 []

scala> counts.cache()
res2: counts.type = ShuffledRDD[12] at reduceByKey at <console>:23

scala> counts.saveAsTextFile("output")
org.apache.hadoop.mapred.FileAlreadyExistsException: Output directory file:/home/student/output already exists
  at org.apache.hadoop.mapred.FileOutputFormat.checkOutputSpecs(FileOutputFormat.java:131)
  at org.apache.spark.internal.io.HadoopMapRedWriteConfigUtil.assertConf(SparkHadoopWriter.scala:299)
  at org.apache.spark.internal.io.SparkHadoopWriter$.write(SparkHadoopWriter.scala:71)
  at org.apache.spark.rdd.PairRDDFunctions.$anonfun$saveAsHadoopDataset$1(PairRDDFunctions.scala:1091)
  at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.java:23)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:151)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:112)
  at org.apache.spark.rdd.RDD.withScope(RDD.scala:410)
  at org.apache.spark.rdd.PairRDDFunctions.saveAsHadoopDataset(PairRDDFunctions.scala:1089)
  at org.apache.spark.rdd.PairRDDFunctions.$anonfun$saveAsHadoopFile$4(PairRDDFunctions.scala:1062)
  at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.java:23)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:151)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:112)
  at org.apache.spark.rdd.RDD.withScope(RDD.scala:410)
  at org.apache.spark.rdd.PairRDDFunctions.saveAsHadoopFile(PairRDDFunctions.scala:1027)
  at org.apache.spark.rdd.PairRDDFunctions.$anonfun$saveAsHadoopFile$3(PairRDDFunctions.scala:1009)
  at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.java:23)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:151)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:112)
  at org.apache.spark.rdd.RDD.withScope(RDD.scala:410)
  at org.apache.spark.rdd.PairRDDFunctions.saveAsHadoopFile(PairRDDFunctions.scala:1008)
  at org.apache.spark.rdd.PairRDDFunctions.$anonfun$saveAsHadoopFile$2(PairRDDFunctions.scala:965)
  at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.java:23)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:151)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:112)
  at org.apache.spark.rdd.RDD.withScope(RDD.scala:410)
  at org.apache.spark.rdd.PairRDDFunctions.saveAsHadoopFile(PairRDDFunctions.scala:963)
  at org.apache.spark.rdd.RDD.$anonfun$saveAsTextFile$2(RDD.scala:1623)
  at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.java:23)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:151)
```

```
student@student:~/TCC48Ass$ cd ..
student@student:~$ cd output/
student@student:~/output$ ls
part-00000  part-00001  _SUCCESS
student@student:~/output$ cat
part-00000       .part-00000.crc  part-00001       .part-00001.crc  _SUCCESS         ._SUCCESS.crc
student@student:~/output$ cat part-00000
(are,1)
(can,1)
(have,1)
(order,1)
(how,3)
(day,1)
(as,1)
(done,1)
(hello,1)
(what,1)
(so,1)
(from,1)
(looks,1)
(artificial,1)
student@student:~/output$ cat part-00001
(it,1)
(inteligence,1)
(was,1)
(a,2)
(text,1)
(be,1)
(u,1)
(written,1)
(cool.,1)
(in,1)
(else,1)
(random,2)
(generated,1)
(doing,1)
(the,1)
student@student:~/output$
```